

27	B	The magnetic force provides the centripetal force on the moving charge. $Bqv = \frac{mv^2}{r} \Rightarrow B = \frac{mv}{qr}$ The proton and the electron carry the same charge q but of opposite sign. For the proton to retrace the path of the electron with the same speed, the radius must be the same. Thus, the magnetic flux density is now $B_p = \frac{m_p v}{-qr}$ $\frac{B_p}{B_e} = -\frac{m_p}{m_e}$ $\frac{B_p}{1.0 \text{ mT}} = -\frac{1.67 \times 10^{-27}}{9.11 \times 10^{-31}}$ $B_p = -1800 \text{ mT}$ The field is in the opposite direction to the initial.
28	C	A. True. As θ from each bit is close that one, the density is constant.